MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCECT301, MG5DSCECC301

- ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A

MCQ

1.(b) To make decisions within a limited time while responding to changing situations

2.(c) Implication

3.(b) The data points that lie exactly on the margin boundaries

4.(a) They help an agent explore possible solutions to reach a goal

5.(c) Reinforcement Learning

6.(a) A state whose neighbors have lower objective function values than the current state

7. (a) A search technique used when an AI agent competes against another agent

8. (a) Fraudulent transactions may differ significantly from normal transactions

9. (a) To know the desired final state

10. (b) Prior distribution

Part B – Short Answer Questions

Answer any 4 questions. Each question carries 5 marks.

11. What is crossover? Explain its role in Genetic Algorithms.

Crossover is a genetic operator that combines genetic information from two parent chromosomes to produce new offspring in a Genetic Algorithm (GA).

Role of crossover:

- Combines useful characteristics of different parents.
- Introduces new combinations of solutions.

- Helps explore the search space.
- May improve the quality of solutions over successive generations.

12. Explain the Naïve Bayes model.

Naïve Bayes is a probabilistic machine-learning classification algorithm based on Bayes' theorem. It assumes that the features are conditionally independent of one another given the class.

Bayes' theorem is:

$$P(C \mid X) = \frac{P(X \mid C)P(C)}{P(X)}$$

Where:

- $P(C \mid X)$: posterior probability of class C.
- $P(X \mid C)$: likelihood of the features given class C.
- $P(C)$: prior probability of class C.
- $P(X)$: evidence or marginal probability of the features.

13. What is the basic difference between Bagging and Boosting?

Bagging	Boosting
Stands for Bootstrap Aggregating.	Combines learners sequentially to improve performance.
Models are trained independently, often in parallel.	Models are trained sequentially, with later learners responding to earlier errors or residuals.
Uses bootstrap samples of the training data.	Commonly changes sample weights or fits residuals, depending on the algorithm.
Combines predictions using voting or averaging.	Combines learners using a weighted or additive prediction.
Primarily reduces variance.	Often reduces bias and can also reduce variance.
Example: Random Forest.	Examples: AdaBoost and Gradient Boosting.

14. Describe the definition and scope of Artificial Intelligence with suitable examples.

Artificial Intelligence (AI) is a branch of computer science concerned with developing systems capable of performing tasks that normally require human intelligence, such as learning, reasoning, problem-solving, perception and decision-making.

Scope of AI:

Machine Learning, Natural Language Processing, Computer Vision, Robotics, Expert Systems, Search and Planning, Probabilistic Reasoning etc.

15. Compare L1 and L2 regularization methods.

Regularization is a technique used in machine learning to reduce overfitting by making the model simpler and improving its performance on new data.

L1 Regularization (Lasso Regression)

- Also called Lasso Regression.
- Adds the absolute value of the coefficients as a penalty to the loss function.
- Shrinks some coefficients exactly to zero, effectively removing less important features.
- Useful for feature selection, especially when there are many features.

L2 Regularization (Ridge Regression)

- Also called Ridge Regression.
- Adds the square of the coefficients as a penalty to the loss function.
- Reduces the size of coefficients but does not make them zero.
- Keeps all features while reducing their influence, helping prevent overfitting.

16. Explain problem formulation in Artificial Intelligence. Describe its important components with a suitable example.

Problem formulation is the process of defining a problem in a structured way so that an AI agent can search for and select actions that lead from an initial situation to a desired goal.

Important components:

1. Initial state: The starting situation of the agent.
2. State space: The set of all possible states.
3. Actions: The operations available to the agent.
4. Transition model: Describes the state reached after performing an action.
5. Goal test: Determines whether the desired state has been reached.
6. Path cost: Measures the total cost of a sequence of actions.

Part C – Essay Questions

Answer any 2 questions. Each question carries 10 marks.

17. Classify independence and conditional independence in probability-based intelligent agents.

- Independence – definition and equation: **4** marks.
- Conditional independence – definition and equation: **4** marks
- Suitable example for each: **2** marks

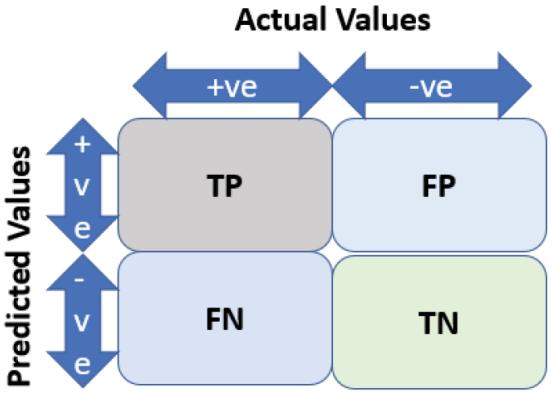
Independence: Two events (A) and (B) are independent if (P(A,B)=P(A)P(B)). The occurrence of one event does not affect the probability of the other.

-Conditional Independence: Two events (A) and (B) are conditionally independent given (C) if (P(A,B| C)=P(A| C)P(B| C)\). Given (C), knowing one event provides no additional information about the other.

18. Construct and illustrate a confusion matrix. Classify and show TP, TN, FP, and FN, and compute the four main evaluation parameters.

- Construction and illustration of confusion matrix – 3 marks
- Identification of TP, TN, FP, and FN – 2 marks
- Four evaluation parameters-3 Marks
- Calculation of Accuracy, Precision, Recall and F1-score – 2 mark

A Confusion Matrix is an N X N matrix used to evaluates the performance of a classification model by comparing actual and predicted values.



Four Evaluation Parameters:

1. Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} \times 100$
2. Precision = $\frac{TP}{TP+FP} \times 100$
3. Recall (Sensitivity) = $\frac{TP}{TP+FN} \times 100$
4. F1-Score = $\frac{2 \times Precision \times Recall}{Precision+Recall}$

19. Explain the A* search algorithm in detail. Describe the role of g(n), h(n), and f(n), and explain how A* selects the next node for expansion.

- Definition of A* search algorithm – 2 marks
- Working principle of A* algorithm – 3 marks
- Explanation of (g(n)), (h(n)), and (f(n)) – 3 marks
- Selection of the next node for expansion – 2 marks

20. Explain Alpha-Beta pruning in detail. Discuss its need, working principle, Alpha and Beta values, and pruning condition.

- Definition of Alpha-Beta pruning – 3 marks
- Need for Alpha-Beta pruning – 1 marks
- Working principle – 2 marks
- Explanation of Alpha (α) and Beta (β) values – 2 marks
- Pruning condition (α≥β) – 2 marks.